

Alexander Gagnon

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Senior SDET with 8+ years of experience building scalable test automation frameworks and ensuring reliability across APIs, web applications, and cloud services. Hands-on architect of Python-based test infrastructure with deep experience in CI/CD integration, AI-augmented workflows, and developer tooling. Passionate about building tools that improve engineering efficiency and software quality.

EXPERIENCE

Manager, QA Automation | Vestmark

Feb 2024 – Present

Team Lead, QA Automation

Feb 2023 – Feb 2024

Senior QA Automation Engineer

Jun 2021 – Feb 2023

- Architected scalable Python test automation frameworks (Selenium, Locust, JMeter, pytest) with integrated CI/CD pipelines (Jenkins, Docker, Gradle), increasing deployment frequency by 40%.
- Led delivery of 700+ automated API tests from the ground up in four months, validating critical data endpoints. Uncovered ~30 previously undetected issues, 5 of them critical.
- Directed development of an AI-powered nightly monitoring tool that automated 2-3 hours of daily manual test triage and validation, saving ~100 hours per month. Built tooling for flake detection and elimination using historical test data. Guided debugging tools for parallel test extraction and DOM analysis, compressing failure turnaround from 2-3 days to same-day.
- Created the QE Marketplace, an enterprise repository of AI-powered plugins (skills, hooks, agents, MCP servers) with training sessions and demos across engineering to drive adoption of automated workflows.
- Led an initiative to redesign test processes through an AI-augmented workflow with human-in-the-loop review and intervention.
- Manage and mentor a team of 11 SDETs across multiple cloud-hosted financial services products. Led a UI automation suite overhaul catching ~120 defects.

Test Automation Software Engineer | Escher Group

Mar 2020 – May 2021

- Standardized Python automation frameworks for desktop and mobile products, reducing production bugs by 10%. Built CI/CD pipelines on Bamboo and Azure Cloud saving 1,000+ developer hours annually. Engineered a full-stack monitoring app with Azure OAuth, cutting developer labor by 500+ hours annually.

Software QA Engineer | Firebrand Technologies

Jul 2019 – Mar 2020

- Joined as a manual tester, identified the absence of automation and performance testing, and self-taught C# and JMeter to independently design, build, and own both frameworks from scratch. Reduced manual effort by 1,200+ hours annually and leveraged performance data to drive infrastructure improvements with leadership. Began training team members to contribute.

Software QA Specialist | Transparent Language

Mar 2018 – Jun 2019

- Promoted from manual tester to database testing. Built Python validation tests for blue/green MongoDB deployments, catching significant data loss during switchovers. Monitored RabbitMQ throughput and identified critical bugs. Earned MongoDB certifications.

PROJECTS

QE Marketplace — Python, Claude Plugins, MCP Servers, Shell & Bash

- Enterprise repository of plugins for bug reporting, test failure triage, documentation generation, and performance goal writing, plus MCP servers for integrations and agents for task execution.

Pilot Regs — Python, RAG, Vector Search

- RAG application giving pilots faster, contextual access to FAA regulations via document ingestion, vector search, and conversational AI.

TECHNICAL SKILLS

Languages & Automation: Python (pytest, unittest), JavaScript/TypeScript, C#, SQL, Bash, Selenium, JMeter, Locust, Groovy

Testing & Frameworks: Test framework architecture, API/REST testing, performance testing, exploratory testing, Jest, root cause analysis

Infrastructure: Jenkins, Docker, Azure, Bamboo, Gradle, Git, GitHub Actions, Grafana, InfluxDB, SonarQube, MongoDB, RabbitMQ

AI & Developer Tooling: Anthropic API, Model Context Protocol (MCP), RAG pipelines, agentic patterns, Claude Desktop, Cursor AI

Collaboration: Technical mentoring, cross-functional collaboration, Scrum/Agile

EDUCATION & MILITARY SERVICE

Bachelor's Degree | University of Massachusetts, Boston

2014 – 2017

Team Leader / Corporal, M1A2 Armor Crewman | United States Army

2010 – 2013